

HE2001 Microeconomics II

Tutorial 3

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Overview

This tutorial focuses on:

- Slutsky vs Hicksian compensation and the relationship between the two;
- utility maximization: marginal utilities, MRS, Marshallian demands, and endowment-income demand;
- net demand and Walras' Law ($p \cdot z(p) = 0$);
- Slutsky income and substitution effects for standard preferences and perfect complements.

1 Slutsky and Hicksian Compensation

A utility maximizing consumer has demand function $x(p_1, p_2, m)$. The price is $p = (p_1, p_2)$. The price of good 1 rises to $p'_1 > p_1$. Show that the Slutsky compensation m_s lies above the Hicksian compensation m_h . That is, show that $m_s \geq m_h$.

Solution

Let the new price vector be $p' = (p'_1, p_2)$. Let the initial chosen bundle be

$$x^0 := x(p_1, p_2, m), \quad u_0 := U(x^0).$$

Define the Slutsky compensation

$$m_s := p' \cdot x^0,$$

and the Hicksian compensation (minimum expenditure at new prices to reach the original utility level)

$$m_h := e(p', u_0) = \min\{p' \cdot y : y \in \mathbb{R}_+^2, U(y) \geq u_0\}.$$

Remark. The phrase “ m_s lies above m_h ” corresponds to $m_s \geq m_h$ (equivalently $m_h \leq m_s$). We prove this inequality below.

Method 1: Feasibility in the expenditure-minimisation problem

By definition,

$$m_h = \min\{p' \cdot y : y \in \mathbb{R}_+^2, U(y) \geq u_0\}.$$

Since $U(x^0) = u_0$, the bundle x^0 is feasible for this minimisation, hence

$$m_h \leq p' \cdot x^0 = m_s.$$

Therefore $m_s \geq m_h$.

$m_s \geq m_h$

Method 2: Indirect utility and the least income needed to reach u_0

Let the indirect utility be

$$v(p, m) := \max\{U(x) : x \in \mathbb{R}_+^2, p \cdot x \leq m\}.$$

Since x^0 is chosen at (p, m) , $v(p, m) = U(x^0) = u_0$. At prices p' , income $m_s = p' \cdot x^0$ makes x^0 affordable, so

$$v(p', m_s) \geq U(x^0) = u_0.$$

Define

$$m_h := \inf\{m' \geq 0 : v(p', m') \geq u_0\}.$$

Because $v(p', m)$ is nondecreasing in m , the inequality $v(p', m_s) \geq u_0$ implies $m_s \geq m_h$.

$m_s \geq m_h$

2 Utility maximization

2.1 Cobb-Douglas Utility and Constant Income Shares

Suppose $U(x_1, x_2) = x_1 x_2$.

1. Calculate the marginal utility for good 1 and 2: MU_1 and MU_2 .
2. Calculate the marginal rate of substitution for goods 1 and 2: MRS_{12} .
3. How much of good 1 will the consumer demand when prices are $(p_1, p_2) = (2, 1)$ and the expenditure is $m = 1$. Hint: use the budget constraint $p_1 x_1 + p_2 x_2 = m$.
4. Calculate the demand function $x(p_1, p_2, m)$.
5. Suppose endowment is $\omega = (3, 1)$. How much will the consumer demand when $(p_1, p_2) = (1, 3)$.

Solution

Method 1: Lagrangian (interior optimum) + plug-in

Maximise $x_1 x_2$ subject to $p_1 x_1 + p_2 x_2 = m$, $x_1, x_2 \geq 0$. Since the objective is strictly increasing on $\mathbb{R}_+^2$ and $m > 0$, the optimum is interior ($x_1, x_2 > 0$) and the budget binds.

Lagrangian:

$$\mathcal{L} = x_1 x_2 + \lambda(m - p_1 x_1 - p_2 x_2).$$

FOCs:

$$\frac{\partial \mathcal{L}}{\partial x_1} = x_2 - \lambda p_1 = 0, \quad \frac{\partial \mathcal{L}}{\partial x_2} = x_1 - \lambda p_2 = 0.$$

From these:

$$\lambda = \frac{x_2}{p_1} = \frac{x_1}{p_2} \Rightarrow \frac{x_2}{x_1} = \frac{p_1}{p_2}.$$

1. Marginal utilities:

$$MU_1 = \frac{\partial}{\partial x_1}(x_1 x_2) = x_2, \quad MU_2 = \frac{\partial}{\partial x_2}(x_1 x_2) = x_1.$$

$$\boxed{MU_1 = x_2, \quad MU_2 = x_1}$$

- 2.

$$MRS_{12} = \frac{MU_1}{MU_2} = \frac{x_2}{x_1}.$$

$$\boxed{MRS_{12} = \frac{x_2}{x_1}}$$

3. For $(p_1, p_2, m) = (2, 1, 1)$, we have $x_2/x_1 = p_1/p_2 = 2$, so $x_2 = 2x_1$. Budget: $2x_1 + x_2 = 1 \Rightarrow 2x_1 + 2x_1 = 1 \Rightarrow x_1 = \frac{1}{4}$.

$$\boxed{x_1 = \frac{1}{4}}$$

4. Using $x_2 = \frac{p_1}{p_2}x_1$ in the budget:

$$p_1x_1 + p_2\left(\frac{p_1}{p_2}x_1\right) = 2p_1x_1 = m \Rightarrow x_1 = \frac{m}{2p_1}, \quad x_2 = \frac{m}{2p_2}.$$

$$x(p_1, p_2, m) = \left(\frac{m}{2p_1}, \frac{m}{2p_2}\right)$$

5. With endowment $\omega = (3, 1)$ and prices $(p_1, p_2) = (1, 3)$, expenditure is

$$m = p \cdot \omega = 1 \cdot 3 + 3 \cdot 1 = 6.$$

Thus

$$x(1, 3, 6) = \left(\frac{6}{2 \cdot 1}, \frac{6}{2 \cdot 3}\right) = (3, 1).$$

$$x = (3, 1)$$

Method 2: MRS = price ratio + budget

At an interior optimum,

$$MRS_{12} = \frac{x_2}{x_1} = \frac{p_1}{p_2} \Rightarrow x_2 = \frac{p_1}{p_2}x_1.$$

Plugging into $p_1x_1 + p_2x_2 = m$ gives $2p_1x_1 = m$, hence $x_1 = m/(2p_1)$ and $x_2 = m/(2p_2)$. All numerical cases follow immediately.

$$x(p_1, p_2, m) = \left(\frac{m}{2p_1}, \frac{m}{2p_2}\right)$$

2.2 Quasilinear Utility and Corner Solutions

Suppose $U(x_1, x_2) = x_1 + 2\sqrt{x_2}$. The marginal utility is $MU_1 = 1$ and $MU_2 = x_2^{-1/2}$.

1. Calculate the marginal rate of substitution for goods 1 and 2: MRS_{12} .
2. How much of good 1 will the consumer demand when prices are $(p_1, p_2) = (2, 1)$ and the expenditure is $m = 1$.
3. Calculate the demand function $x(p_1, p_2, m)$.
4. Suppose endowment is $\omega = (3, 1)$. How much will the consumer demand when $(p_1, p_2) = (2, 1)$.

Solution

Method 1: Tangency candidate + corner verification

1. Since $MU_1 = 1$ and $MU_2 = x_2^{-1/2} = 1/\sqrt{x_2}$,

$$MRS_{12} = \frac{MU_1}{MU_2} = \frac{1}{1/\sqrt{x_2}} = \sqrt{x_2}.$$

$$\boxed{MRS_{12} = \sqrt{x_2}}$$

2. At an interior optimum, tangency requires $MRS_{12} = p_1/p_2$, i.e.

$$\sqrt{x_2} = \frac{p_1}{p_2}.$$

At $(p_1, p_2) = (2, 1)$, this gives $x_2 = 4$, which is infeasible under the budget $2x_1 + x_2 \leq 1$. Hence the optimum is a corner.

Since U is increasing in both goods, the budget binds: $2x_1 + x_2 = 1$. Write $x_1 = (1 - x_2)/2$ with $x_2 \in [0, 1]$, and maximise

$$f(x_2) = \frac{1 - x_2}{2} + 2\sqrt{x_2}.$$

Derivative for $x_2 \in (0, 1]$:

$$f'(x_2) = -\frac{1}{2} + \frac{1}{\sqrt{x_2}}.$$

For $x_2 \in (0, 1]$, $1/\sqrt{x_2} \geq 1$, so $f'(x_2) \geq \frac{1}{2} > 0$. Thus f is increasing, and the maximiser is $x_2^* = 1$, hence $x_1^* = 0$.

$$\boxed{x_1 = 0}$$

3. Interior candidate from tangency:

$$\sqrt{x_2} = \frac{p_1}{p_2} \Rightarrow x_2^\circ = \left(\frac{p_1}{p_2}\right)^2, \quad x_1^\circ = \frac{m - p_2 x_2^\circ}{p_1} = \frac{m}{p_1} - \frac{p_1}{p_2}.$$

This is feasible with $x_1^o \geq 0$ iff $m \geq p_1^2/p_2$. If $m < p_1^2/p_2$, the optimum is the corner $x_1^* = 0$, $x_2^* = m/p_2$.

$$x(p_1, p_2, m) = \begin{cases} \left(\frac{m}{p_1} - \frac{p_1}{p_2}, \left(\frac{p_1}{p_2} \right)^2 \right), & m \geq \frac{p_1^2}{p_2}, \\ \left(0, \frac{m}{p_2} \right), & m < \frac{p_1^2}{p_2}. \end{cases}$$

4. With endowment $\omega = (3, 1)$ and prices $(p_1, p_2) = (2, 1)$,

$$m = p \cdot \omega = 2 \cdot 3 + 1 \cdot 1 = 7.$$

Since $7 \geq 2^2/1 = 4$, the interior case applies:

$$x_2 = \left(\frac{2}{1} \right)^2 = 4, \quad x_1 = \frac{7}{2} - \frac{2}{1} = \frac{3}{2}.$$

$$x = \left(\frac{3}{2}, 4 \right)$$

Method 2: One-dimensional maximisation on the budget set (for the corner case)

For $(p_1, p_2, m) = (2, 1, 1)$, the feasible set is $\{(x_1, x_2) : 2x_1 + x_2 \leq 1, x_1, x_2 \geq 0\}$. Because U is increasing, the constraint binds, so $x_1 = (1 - x_2)/2$ with $x_2 \in [0, 1]$. Define

$$f(x_2) = U\left(\frac{1 - x_2}{2}, x_2\right) = \frac{1 - x_2}{2} + 2\sqrt{x_2}.$$

As computed above,

$$f'(x_2) = -\frac{1}{2} + \frac{1}{\sqrt{x_2}} > 0 \quad \text{for all } x_2 \in (0, 1],$$

so the maximiser is $x_2^* = 1$, hence $x_1^* = 0$. Therefore $x_1(2, 1, 1) = 0$.

$$x_1(2, 1, 1) = 0$$

3 Demand example

Consider the utility function

$$U(x) = \sqrt{x_1} + \sqrt{x_2}.$$

This U generates the demand function

$$x(p_1, p_2, m) = \frac{m}{p_1 + p_2} \left(\frac{p_2}{p_1}, \frac{p_1}{p_2} \right).$$

Suppose the consumer with utility function U has endowment $\omega = (3, 1)$.

1. Show that $p \cdot z(p) = 0$ where z is the net demand function (i.e. $z(p) = x(p, p \cdot \omega) - \omega$).
2. What is the consumer's net demand $z(p)$ at $p = (1, 1)$? Is the consumer a net buyer or seller of good 1?
3. Prices change to $p' = (2, 1)$. What is the net demand $z(p')$ at the new price level?
4. Calculate the income and substitution effects for good 1 for this price change.

Solution

Method 1: Direct calculation

Let $m(p) := p \cdot \omega$. Then $z(p) = x(p, m(p)) - \omega$.

1. Since the demand exhausts the budget,

$$p \cdot x(p, m) = m.$$

Setting $m = m(p) = p \cdot \omega$ gives $p \cdot x(p, p \cdot \omega) = p \cdot \omega$. Hence

$$p \cdot z(p) = p \cdot (x(p, p \cdot \omega) - \omega) = 0.$$

$$\boxed{p \cdot z(p) = 0}$$

2. At $p = (1, 1)$, income is $m = (1, 1) \cdot (3, 1) = 4$. Then

$$x(1, 1, 4) = \frac{4}{2}(1, 1) = (2, 2),$$

so

$$z(1, 1) = (2, 2) - (3, 1) = (-1, 1).$$

Thus $z_1(1, 1) = -1 < 0$, so the consumer is a net seller of good 1.

$$\boxed{z(1, 1) = (-1, 1) \text{ and net seller of good 1.}}$$

3. At $p' = (2, 1)$, income is $m' = (2, 1) \cdot (3, 1) = 7$. Then

$$x(2, 1, 7) = \frac{7}{3} \left(\frac{1}{2}, 2 \right) = \left(\frac{7}{6}, \frac{14}{3} \right),$$

so

$$z(2, 1) = \left(\frac{7}{6}, \frac{14}{3} \right) - (3, 1) = \left(-\frac{11}{6}, \frac{11}{3} \right).$$

$$\boxed{z(2, 1) = \left(-\frac{11}{6}, \frac{11}{3} \right)}$$

4. Initial prices $p = (1, 1)$, initial income $m_0 = 4$, initial bundle $x^0 = x(p, m_0) = (2, 2)$, so $x_1^0 = 2$. New prices $p' = (2, 1)$, new income $m_1 = 7$, new bundle $x^1 = x(p', m_1) = (7/6, 14/3)$, so $x_1^1 = 7/6$.

Slutsky (compensated) income:

$$m_s := p' \cdot x^0 = (2, 1) \cdot (2, 2) = 6.$$

Compensated bundle:

$$x^c := x(p', m_s) = x(2, 1, 6) = \frac{6}{3} \left(\frac{1}{2}, 2 \right) = (1, 4),$$

so $x_1^c = 1$. Therefore

$$\Delta x_1^s := x_1^c - x_1^0 = 1 - 2 = -1, \quad \Delta x_1^n := x_1^1 - x_1^c = \frac{7}{6} - 1 = \frac{1}{6}.$$

$$\boxed{\Delta x_1^s = -1, \quad \Delta x_1^n = \frac{1}{6}}$$

Method 2: Three-point evaluation of x_1

Compute x_1 at three points:

$$x_1(p, m_0) = 2, \quad x_1(p', m_s) = 1, \quad x_1(p', m_1) = \frac{7}{6}.$$

Then

$$\Delta x_1^s = 1 - 2 = -1, \quad \Delta x_1^n = \frac{7}{6} - 1 = \frac{1}{6}.$$

$$\boxed{\Delta x_1^s = -1, \quad \Delta x_1^n = \frac{1}{6}}$$

4 Perfect Complements

Consider the utility function

$$U(x_1, x_2) = \min\{x_1, x_2\}.$$

So, $U(2, 3) = 2$, $U(6, 1) = 1$, $U(100, 3) = 3$ and so on. The demand function is

$$x(p_1, p_2, m) = \left(\frac{m}{p_1 + p_2}, \frac{m}{p_1 + p_2} \right),$$

and the consumer has endowment $\omega = (3, 1)$.

1. Show that $p \cdot z(p) = 0$ where z is the net demand function (i.e. $z(p) = x(p, p \cdot \omega) - \omega$).
2. What is the consumer's net demand $z(p)$ at $p = (1, 1)$? Is the consumer a net buyer or seller of good 1?
3. Prices change to $p' = (2, 1)$. What is the net demand $z(p')$ at the new price level?
4. Calculate the income and substitution effects for good 1 for this price change.

Solution

Method 1: Direct calculation from the demand function

Let $m(p) := p \cdot \omega$. Then $z(p) = x(p, m(p)) - \omega$.

1. For any m ,

$$p \cdot x(p, m) = p_1 \frac{m}{p_1 + p_2} + p_2 \frac{m}{p_1 + p_2} = m.$$

Setting $m = p \cdot \omega$ gives $p \cdot x(p, p \cdot \omega) = p \cdot \omega$, hence $p \cdot z(p) = 0$.

$$\boxed{p \cdot z(p) = 0}$$

2. At $p = (1, 1)$, income is $m = (1, 1) \cdot (3, 1) = 4$. Demand:

$$x(1, 1, 4) = \left(\frac{4}{2}, \frac{4}{2} \right) = (2, 2).$$

Net demand:

$$z(1, 1) = (2, 2) - (3, 1) = (-1, 1).$$

Thus $z_1(1, 1) = -1 < 0$, so the consumer is a net seller of good 1.

$$\boxed{z(1, 1) = (-1, 1) \text{ and net seller of good 1.}}$$

3. At $p' = (2, 1)$, income is $m' = (2, 1) \cdot (3, 1) = 7$. Demand:

$$x(2, 1, 7) = \left(\frac{7}{3}, \frac{7}{3}\right).$$

Net demand:

$$z(2, 1) = \left(\frac{7}{3}, \frac{7}{3}\right) - (3, 1) = \left(-\frac{2}{3}, \frac{4}{3}\right).$$

$$\boxed{z(2, 1) = \left(-\frac{2}{3}, \frac{4}{3}\right)}$$

4. Initial prices $p = (1, 1)$, initial income $m_0 = 4$, initial bundle $x^0 = (2, 2)$, so $x_1^0 = 2$. New prices $p' = (2, 1)$, new income $m_1 = 7$, new bundle $x^1 = (7/3, 7/3)$, so $x_1^1 = 7/3$.

Slutsky income:

$$m_s := p' \cdot x^0 = (2, 1) \cdot (2, 2) = 6.$$

Compensated bundle:

$$x^c := x(p', m_s) = x(2, 1, 6) = \left(\frac{6}{3}, \frac{6}{3}\right) = (2, 2),$$

so $x_1^c = 2$. Therefore

$$\Delta x_1^s := x_1^c - x_1^0 = 2 - 2 = 0, \quad \Delta x_1^m := x_1^1 - x_1^c = \frac{7}{3} - 2 = \frac{1}{3}.$$

$$\boxed{\Delta x_1^s = 0, \quad \Delta x_1^m = \frac{1}{3}}$$

Method 2: Perfect complements imply zero substitution effect

With $U(x_1, x_2) = \min\{x_1, x_2\}$, utility depends only on matched pairs, so the compensated (utility-held-constant) bundle does not move away from the kink. For the initial bundle $x^0 = (2, 2)$, the relevant kink level is 2, and the least-cost way to achieve this level at any prices is still to consume equal quantities. Thus the compensated demand at $p' = (2, 1)$ remains $(2, 2)$, giving $\Delta x_1^s = 0$. The remaining change from 2 to $7/3$ is the income effect $1/3$.

$$\boxed{\Delta x_1^s = 0, \quad \Delta x_1^m = \frac{1}{3}}$$